

6(a)	$V = V_1 + V_2 + V_3$	B1
	$IR = IR_1 + IR_2 + IR_3$ or $(V/I) = (V_1/I) + (V_2/I) + (V_3/I)$	B1
	and $R = R_1 + R_2 + R_3$	
6(b)(i)	$V/8.0 = 6.0 \times 10^3 / (4.0 \times 10^3 + 6.0 \times 10^3)$	C1
	or $I = 8.0 / (4.0 \times 10^3 + 6.0 \times 10^3) = 8.0 \times 10^{-4}$ $V = 8.0 \times 10^{-4} \times 6.0 \times 10^3$	
	$V = 4.8 \text{ V}$	A1
6(b)(ii)	total resistance in parallel = $3.0 \times 10^3 (\Omega)$ or $3.0 (\text{k}\Omega)$	C1
	current = $8.0 / (3.0 \times 10^3 + 6.0 \times 10^3)$ $= 8.9 \times 10^{-4} \text{ A}$	A1
6(c)	<u>thermistor</u> resistance increases	B1
	(thermistor resistance increases so total resistance increases so) current decreases (in battery)	M1
	($P = EI$ and E constant so) power decreases	A1